

Exploring Aligned and Disaligned Multimodal Features in Recommender Systems

Emanuele Fontana

December 23, 2025

Outline

- 1 Introduction
- 2 Aligned Models
- 3 Experimental Settings
- 4 Experiments
- 5 Discussion & Conclusion

Outline

- 1 Introduction
- 2 Aligned Models
- 3 Experimental Settings
- 4 Experiments
- 5 Discussion & Conclusion

Recommender Systems

- Guide users to relevant items in a large space [5]
- Approaches: Content-Based, Collaborative, Hybrid, Knowledge-Based, Context-Aware

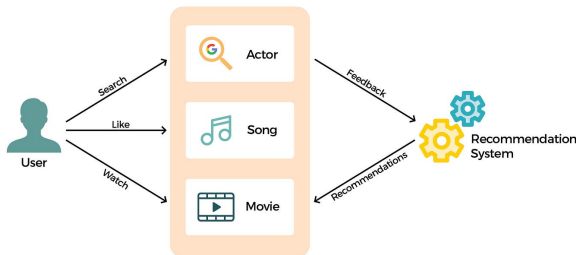


Figure: Example of a Recommender System

Multimodal Recommender Systems

- Use multiple modalities (text, image, audio) to enrich item representations [7]
- Encoding options: concat, fusion/attention, GNNs [1, 4]

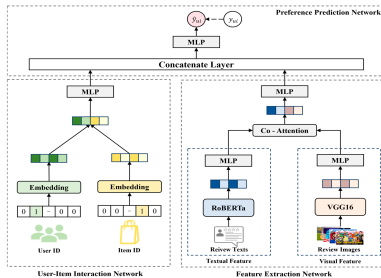


Figure: Example of a Multimodal Recommender System

Outline

- 1 Introduction
- 2 Aligned Models**
- 3 Experimental Settings
- 4 Experiments
- 5 Discussion & Conclusion

Contrastive Learning

- Self-supervised method learning representations that bring similar samples closer [3]
- Contrastive loss: $\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} [k \neq i] \exp(\text{sim}(z_i, z_k)/\tau)}$

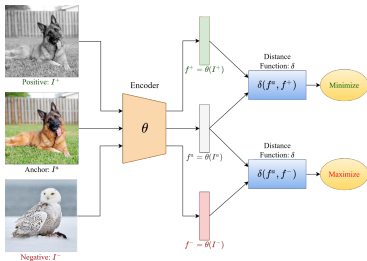


Figure: Conceptual representation of Contrastive Learning

- CLIP aligns image and text embeddings in a shared space (OpenAI) [6]
- Recent work shows that large multimodal encoders like CLIP can effectively improve recommender system performance when fine-tuned appropriately [8]

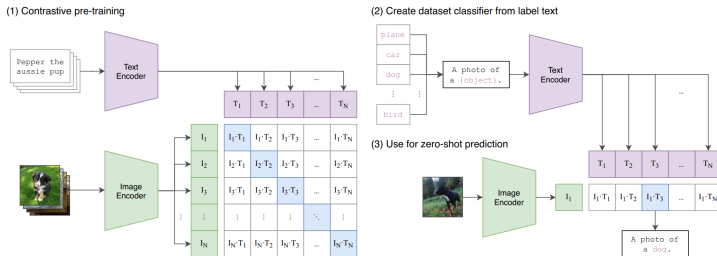


Figure: Conceptual representation of CLIP architecture

AudioCLIP

- Extends CLIP by adding an audio encoder (ESResNeXt) to the tri-modal alignment [2]
- Training: audio-head pretrain, cooperative distillation, full tri-modal fine-tuning on AudioSet

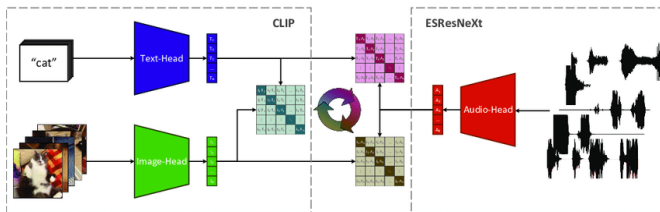


Figure: Conceptual representation of AudioCLIP architecture

Outline

- 1 Introduction
- 2 Aligned Models
- 3 Experimental Settings**
- 4 Experiments
- 5 Discussion & Conclusion

Research Questions, Datasets and Models

RQ1: Can aligned multimodal embeddings improve recommender performance? Considered dataset:

- MovieLens 1M: movies with posters, trailers (audio), and text descriptions
- LastFM: artist-level interactions, track-level multimodal features (images, audio, text)

Considered models:

- BPR: Bayesian Personalized Ranking
- VBPR: extension of BPR incorporating for multimodal recommendation
- LATTICE: GNN-based model for multimodal recommendation (with 1 or 3 layers)
- FREEDOM: GNN-based model with modality-specific encoders and attention-based fusion (3 layers)

Outline

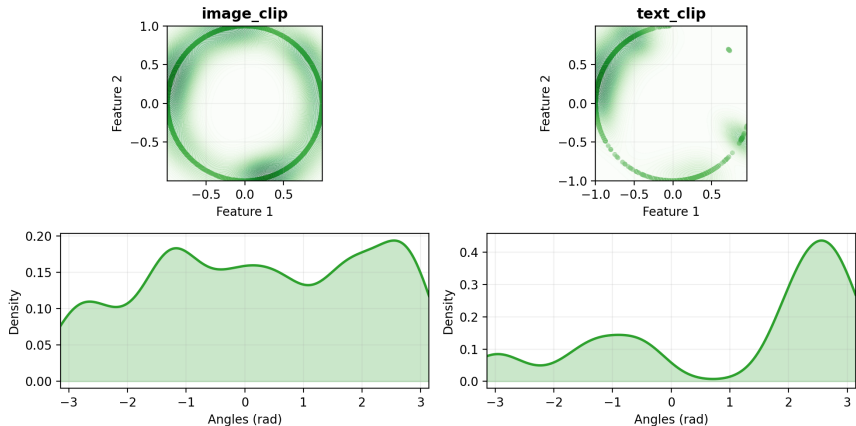
- 1 Introduction
- 2 Aligned Models
- 3 Experimental Settings
- 4 Experiments**
- 5 Discussion & Conclusion

Embedding distribution technique

- UMAP (Uniform Manifold Approximation and Projection) is used to reduce high-dimensional embeddings to 2D preserving local/global structure.
- After UMAP, we center and S1-normalize points: $\mathbf{x}'_i = \mathbf{x}_i - \bar{\mathbf{x}}$,
 $\mathbf{x}''_i = \frac{\mathbf{x}'_i}{\|\mathbf{x}'_i\|}$.
- Spatial density: 2D Gaussian Kernel Density Estimation (KDE) on the projected points.

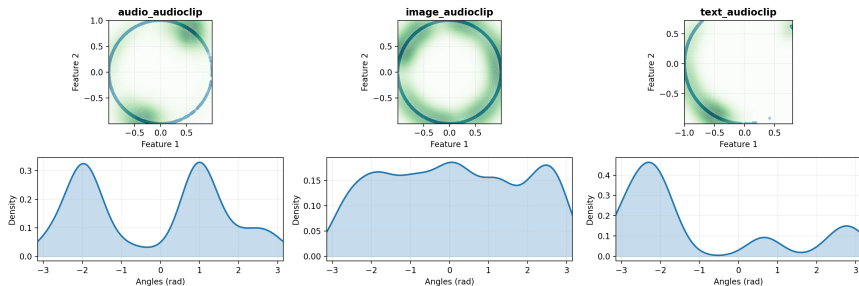
MovieLens 1M: CLIP-aligned Text+Image

MOVIELENS_1M - text_clip_image_clip



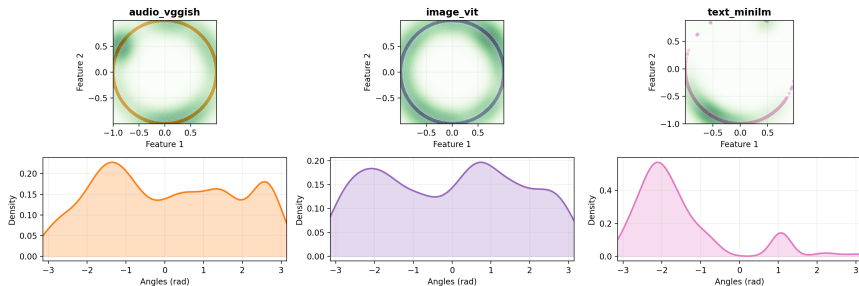
MovieLens 1M: AudioCLIP aligned (Text+Image+Audio)

MOVIELENS_1M - audioclip_full



MovieLens 1M: Disaligned Text+Image+Audio (MiniLM+ViT+VGGish)

MOVIELENS_1M - text_minilm_image_vit_audio_vggish



Quantitative comparison: MovieLens (k=5)

Model	Rec@5	NDCG@5	Prec@5	MAP@5
BPR	0.1089	0.2096	0.1898	0.1289
VBPR (CLIP)	0.1152	0.2205	0.1997	0.1376
VBPR (AudioCLIP)	0.0940	0.1857	0.1694	0.1125
VBPR (Disaligned)	0.1163	0.2191	0.1986	0.1356
LATTICE_3 (CLIP)	0.1134	0.2189	0.1977	0.1359
LATTICE_3 (AudioCLIP)	0.1120	0.2183	0.1988	0.1358
LATTICE_3 (Disaligned)	0.1132	0.2179	0.1971	0.1349
FREEDOM (CLIP)	0.1106	0.2150	0.1930	0.1331
FREEDOM (AudioCLIP)	0.1164	0.2204	0.1984	0.1373
FREEDOM (Disaligned)	0.1106	0.2158	0.1938	0.1337

Table: MovieLens 1M - Comparison at cutoff k=5

Quantitative comparison: MovieLens (k=10)

Model	Rec@10	NDCG@10	Prec@10	MAP@10
BPR	0.1776	0.2159	0.1631	0.1120
VBPR (CLIP)	0.1911	0.2300	0.1745	0.1211
VBPR (AudioCLIP)	0.1570	0.1922	0.1476	0.0969
VBPR (Disaligned)	0.1876	0.2271	0.1718	0.1195
LATTICE_3 (CLIP)	0.1877	0.2275	0.1721	0.1198
LATTICE_3 (AudioCLIP)	0.1875	0.2266	0.1725	0.1188
LATTICE_3 (Disaligned)	0.1848	0.2257	0.1710	0.1183
FREEDOM (CLIP)	0.1796	0.2211	0.1657	0.1158
FREEDOM (AudioCLIP)	0.1844	0.2259	0.1688	0.1193
FREEDOM (Disaligned)	0.1801	0.2215	0.1658	0.1161

Table: MovieLens 1M - Comparison at cutoff k=10

Quantitative comparison: LastFM (k=5)

Model	Rec@5	NDCG@5	Prec@5	MAP@5
BPR	0.0655	0.1595	0.1577	0.1414
VBPR (CLIP)	0.0707	0.1714	0.1691	0.1524
VBPR (AudioCLIP)	0.0694	0.1735	0.1711	0.1531
VBPR (Disaligned)	0.0768	0.1859	0.1831	0.1660
LATTICE_3 (CLIP)	0.0726	0.1762	0.1751	0.1564
LATTICE_3 (AudioCLIP)	0.0712	0.1742	0.1718	0.1538
LATTICE_3 (Disaligned)	0.0757	0.1856	0.1825	0.1646
FREEDOM (CLIP)	0.0522	0.1357	0.1358	0.1303
FREEDOM (AudioCLIP)	0.0526	0.1358	0.1358	0.1299
FREEDOM (Disaligned)	0.0527	0.1362	0.1362	0.1306

Table: LastFM - Comparison at cutoff k=5

Quantitative comparison: LastFM (k=10)

Model	Rec@10	NDCG@10	Prec@10	MAP@10
BPR	0.1132	0.1534	0.1404	0.1061
VBPR (CLIP)	0.1222	0.1646	0.1488	0.1149
VBPR (AudioCLIP)	0.1196	0.1636	0.1482	0.1136
VBPR (Disaligned)	0.1307	0.1755	0.1560	0.1242
LATTICE_3 (CLIP)	0.1232	0.1667	0.1507	0.1160
LATTICE_3 (AudioCLIP)	0.1238	0.1669	0.1510	0.1160
LATTICE_3 (Disaligned)	0.1258	0.1722	0.1535	0.1209
FREEDOM (CLIP)	0.0986	0.1320	0.1225	0.0953
FREEDOM (AudioCLIP)	0.0989	0.1323	0.1227	0.0953
FREEDOM (Disaligned)	0.0986	0.1322	0.1224	0.0954

Table: LastFM - Comparison at cutoff k=10

Outline

- 1 Introduction
- 2 Aligned Models
- 3 Experimental Settings
- 4 Experiments
- 5 Discussion & Conclusion**

- Alignment helps when modalities share strong semantic relationships (e.g., posters and descriptions)
- Aggressive alignment (AudioCLIP) can erase modality-specific signals leading to worse ranking
- Simpler models (VBPR) can outperform complex GNNs when features are well represented

Conclusions and Future Work

- Use domain-adaptive fusion strategies to select aligned/disaligned features
- Explore softer alignment / regularization strategies for audio
- Investigate model-feature co-design (when to use GNNs vs. projection models)

Thank you!

Thank you for your attention! 🚀

References I



Xi Chen, Yangsiyi Lu, Yuehai Wang, and Jianyi Yang.
Cmbf: Cross-modal-based fusion recommendation algorithm.
Sensors, 21(16), 2021.



Andrey Guzhov, Federico Raue, Jörn Hees, and Andreas Dengel.
Audioclip: Extending clip to image, text and audio.
In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 976–980. IEEE, 2022.



Haigen Hu, Xiaoyuan Wang, Yan Zhang, Qi Chen, and Qiu Guan.
A comprehensive survey on contrastive learning.
Neurocomputing, 610:128645, 2024.

References II



Peishan Li, Weixiao Zhan, Lutao Gao, Shuran Wang, and Linnan Yang.

Multimodal recommendation system based on cross self-attention fusion.

Systems, 13(1), 2025.



Pasquale Lops, Marco de Gemmis, and Giovanni Semeraro.

Content-based Recommender Systems: State of the Art and Trends, pages 73–105.

Springer US, Boston, MA, 2011.

References III



Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al.

Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.



Giuseppe Spillo, Elio Musacchio, Cataldo Musto, Marco de Gemmis, Pasquale Lops, and Giovanni Semeraro.

See the movie, hear the song, read the book: Extending movielens-1m, last.fm-2k, and dbbook with multimodal data.

In *Proceedings of the Nineteenth ACM Conference on Recommender Systems, RecSys '25*, page 847–856, New York, NY, USA, 2025. Association for Computing Machinery.



Zixuan Yi, Zijun Long, Iadh Ounis, Craig Macdonald, and Richard Mccreadie.

Large multi-modal encoders for recommendation.

arXiv preprint arXiv:2310.20343, 2023.